

Atlantic Hurricane Tracks: an exploration of HURDAT2

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Contents

Introduction	1
Data	1
Part 1: dataset summaries	2
Part 2: spatial visualisation	2
Part 3: storms near a reference city	6
Part 4: temporal patterns	6
Discussion	6
Reproducibility	13

Introduction

This report explores the NOAA **HURDAT2** Atlantic-basin hurricane dataset. HURDAT2 is the official best-track archive for tropical cyclones in the North Atlantic; the version used here covers 1851–2025 and contains roughly 2,000 storms and 56,000 track points. Each row is a single six-hour observation of a storm: position, intensity, and (for landfalling points) a marker. We use it to answer four broad questions:

1. What does the dataset look like in aggregate?
2. What does the spatial distribution of storm tracks look like?
3. Which storms have passed close to a chosen reference city?
4. How have storm frequency, intensity, and season length changed over time?

The analysis is an updated, cleaned-up version of a 2023 coursework project. A few bugs in the original draft (a latitude-bounds check that was silently dead code, an unvectorised distance function being called on a vector, a malformed loop accumulator) have been fixed; the methodology is otherwise unchanged.

Data

Daily observations are loaded from a tidy snapshot produced by `data/fetch_data.R`. The script downloads the current HURDAT2 file from `nhc.noaa.gov`, parses the fixed-width format, and saves a tidy data frame called `hurrs` with one row per observation:

Table 1: HURDAT2 dataset summary

Statistic	Value
Storms	2004
Track points	55605
First year	1851
Last year	2025
Landfall points	1175
Observations with wind data	55548

Part 1: dataset summaries

We can pull out, for example, the full track of Hurricane Katrina (2005):

```
katrina <- hurrs %>% filter(id == "AL122005")
cat(sprintf("Katrina (AL122005) track points: %d\n", nrow(katrina)))
```

```
## Katrina (AL122005) track points: 34
```

A per-storm summary function is useful for describing any subset:

```
Hurdat2_summary <- function(x) {
  list(
    storms      = n_distinct(x$id),
    year_span   = max(x$year) - min(x$year),
    track_points = x %>% group_by(id) %>% summarise(points = n())
  )
}
```

Storms are named from 1950 onward:

```
hurrs %>% filter(name != "UNNAMED") %>% arrange(year) %>% head(1) %>% select(id, name, year)
```

```
##           id name year
## 1 AL011950 ABLE 1950
```

Part 2: spatial visualisation

A reusable basemap for the Atlantic basin, with overlay tracks coloured by windspeed:

```
hurricaneBasemap <- function() {
  world <- map_data("world")
  ggplot() +
    coord_fixed(xlim = c(-130, 30), ylim = c(0, 90)) +
    geom_polygon(data = world, aes(x = long, y = lat, group = group),
                color = "gray50", fill = "white")
}
```

A single-year view (Hurricane Andrew and its peers in 1992):

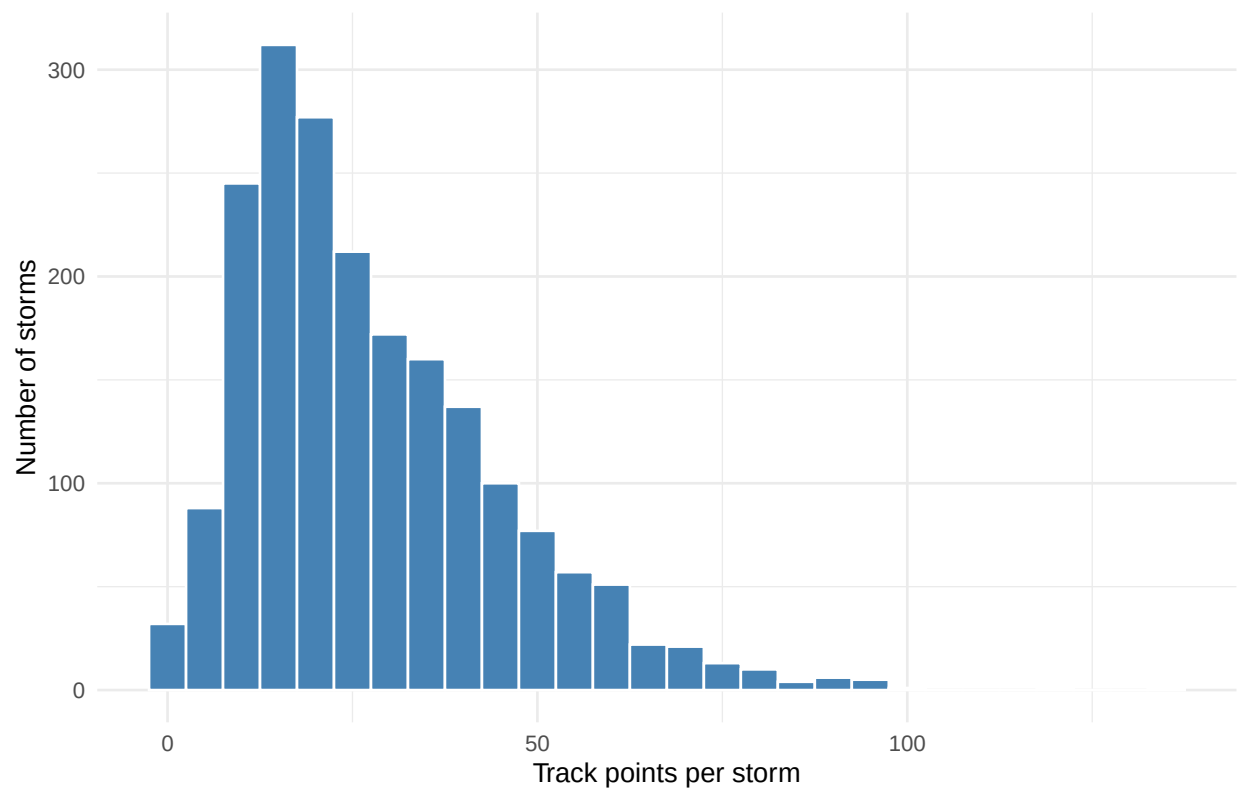


Figure 1: Distribution of track-point counts per storm. The long right tail reflects the longest-lived storms.

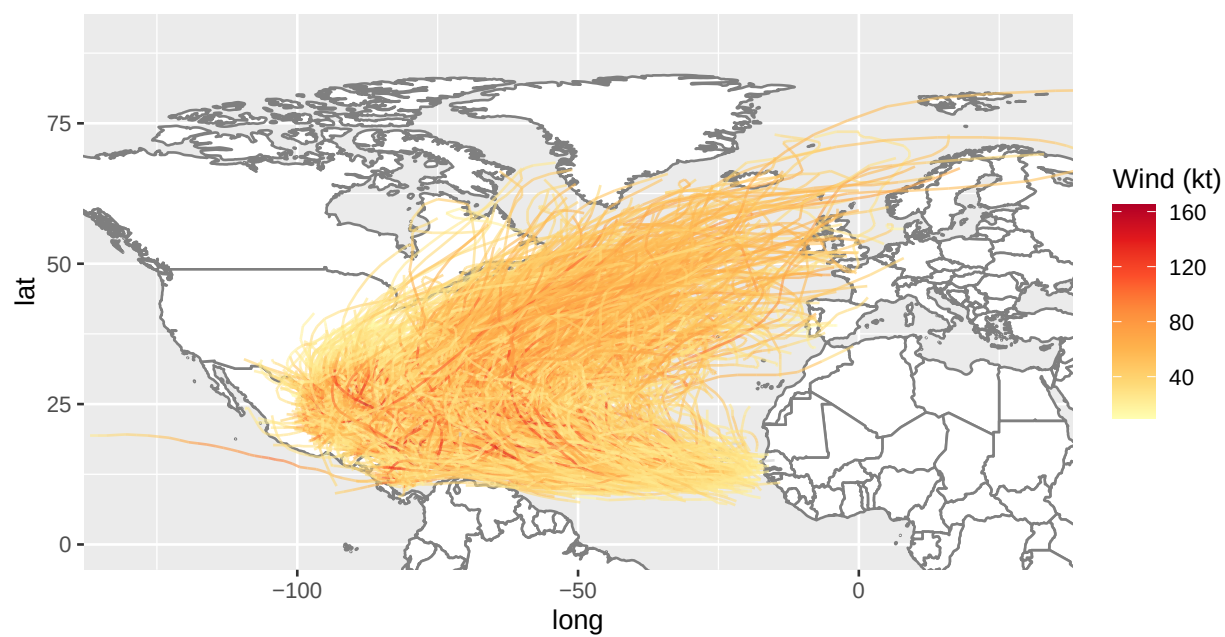


Figure 2: All recorded Atlantic storm tracks, coloured by windspeed. Pre-1900 storms lack wind data and appear grey.

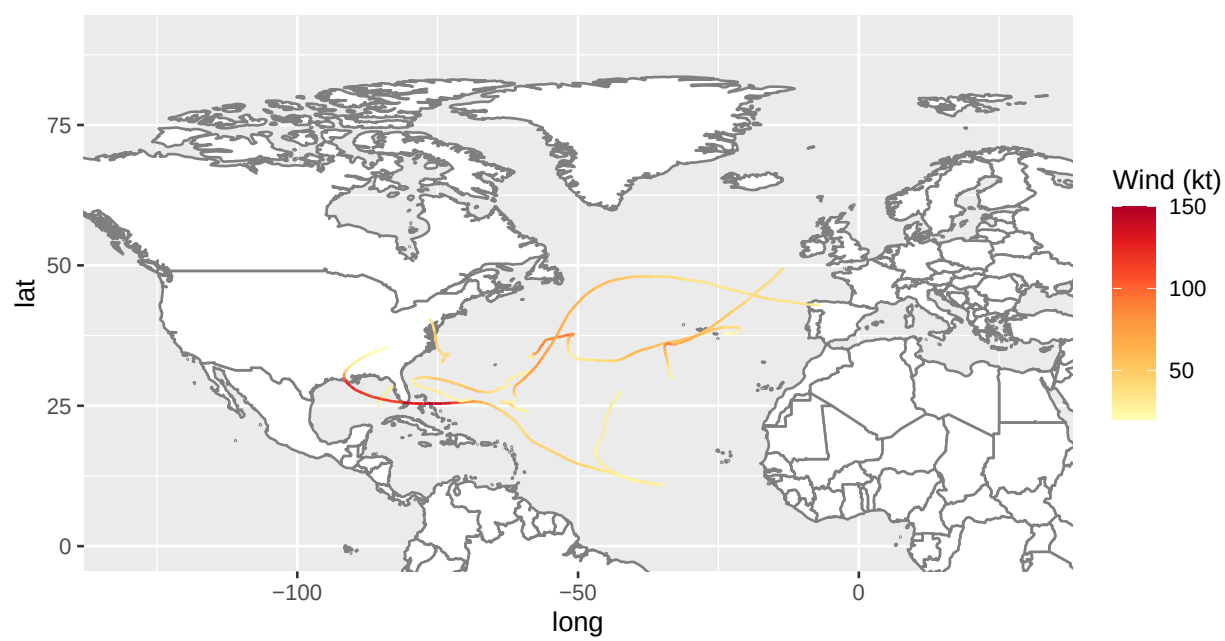


Figure 3: Storm tracks in 1992, coloured by windspeed.

Part 3: storms near a reference city

To make distance questions tractable we use an equirectangular approximation of great-circle distance. Because the function is called inside a `mutate` on a 56,000-row data frame, it has to be vectorised — the original draft called a scalar implementation on a vector and silently mis-behaved.

```
findD <- function(xx, yy, x, y) {
  dy <- yy - y
  dx <- (xx - x) * cos((pi / 180) * ((y + yy) / 2))
  111.325 * sqrt(dx^2 + dy^2)
}

findD_city <- function(x, y, city) {
  ref <- world.cities %>% filter(name == city) %>% slice(1)
  if (nrow(ref) == 0) stop("city not found: ", city)
  findD(ref$long, ref$lat, x, y)
}

# v2 adds a sanity check for swapped lon/lat inputs. The original draft had
# `if (y <-90 & y> 90)` which R parses as `y <- 90; (90 > 90)` - a no-op.
findD_city_v2 <- function(x, y, city) {
  if (any(y < -90 | y > 90, na.rm = TRUE)) {
    warning("Latitude outside [-90, 90]: check for swapped lon/lat.")
  }
  findD_city(x, y, city)
}
```

Which storms have passed within 100 km of Port-au-Prince?

Part 4: temporal patterns

Storms per year

The steep rise in the second half of the 20th century is at least partly an artefact of better observation — satellites began tracking the open Atlantic in the 1960s, so storms that would have gone unrecorded earlier now get a best-track entry.

Landfall zone over time

Peak windspeed per year

Storms per month (the hurricane season)

Season expansion across decades

Stacking each year's storms by month shows the season widening over time: recent decades have seen off-season storms (January, March, April) that were essentially absent in the 19th-century record.

Discussion

A few takeaways:

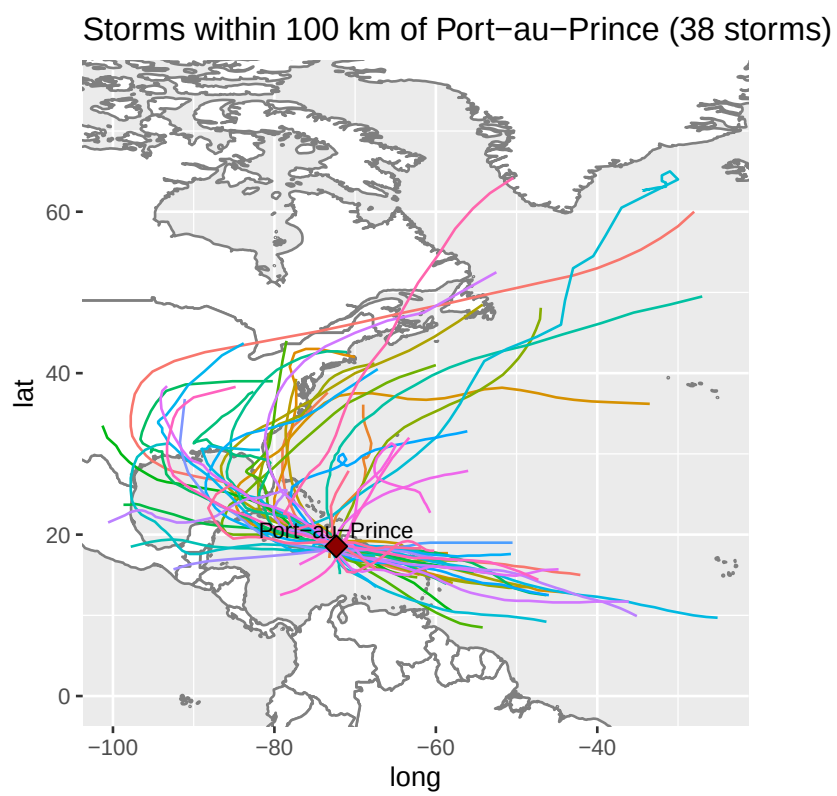


Figure 4: Storm tracks that have passed within 100 km of Port-au-Prince.

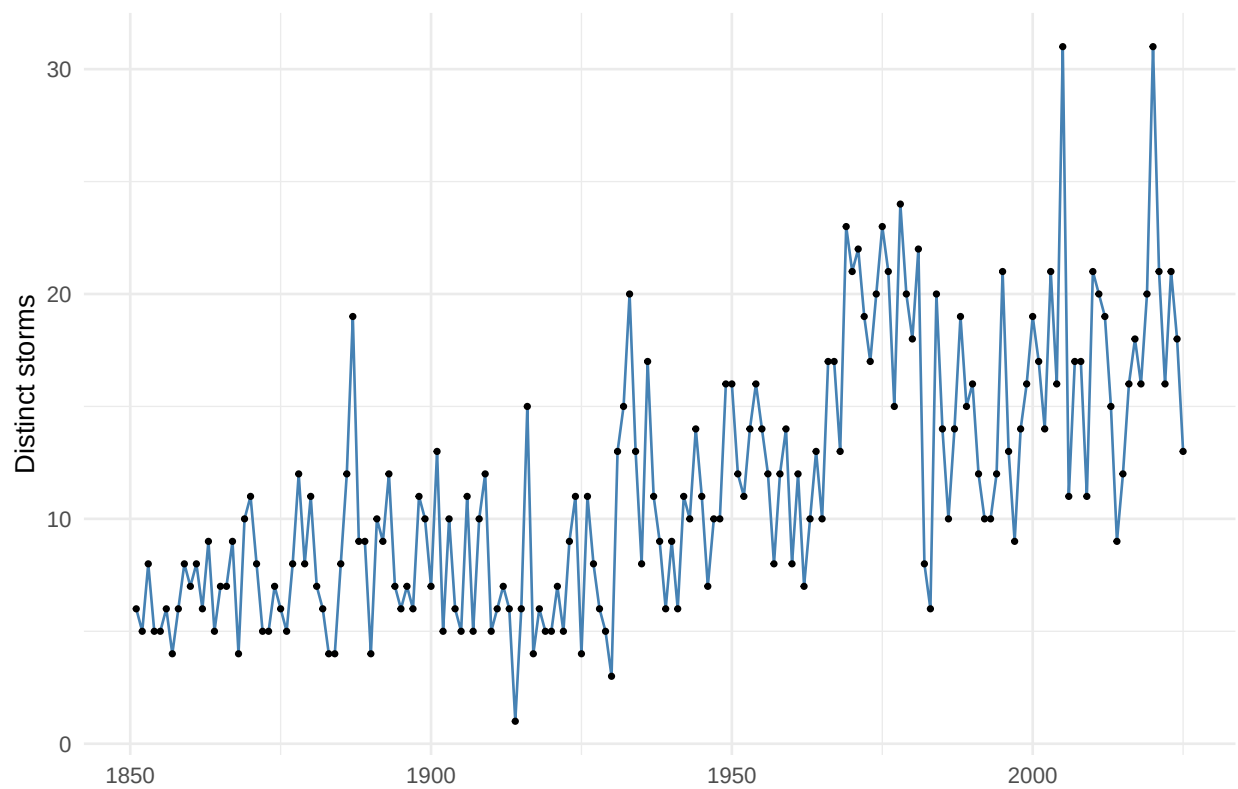


Figure 5: Distinct Atlantic storms recorded per year. Reporting density is much lower before the satellite era (post-1960s).

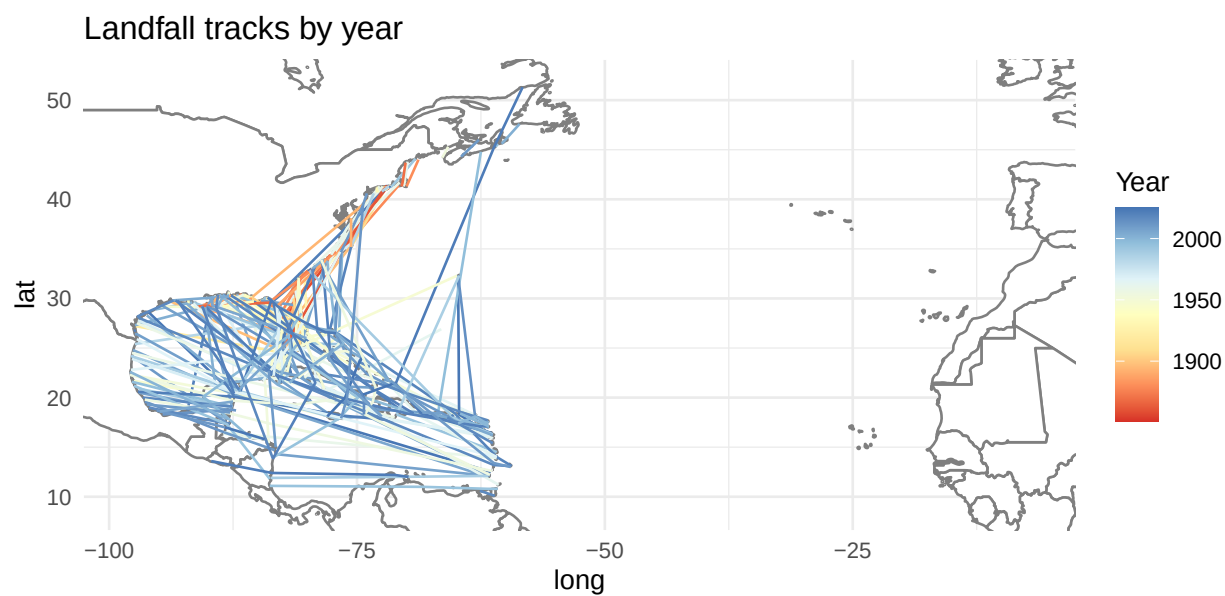


Figure 6: Landfall tracks coloured by year. The set of coastlines affected has expanded poleward and westward over time.

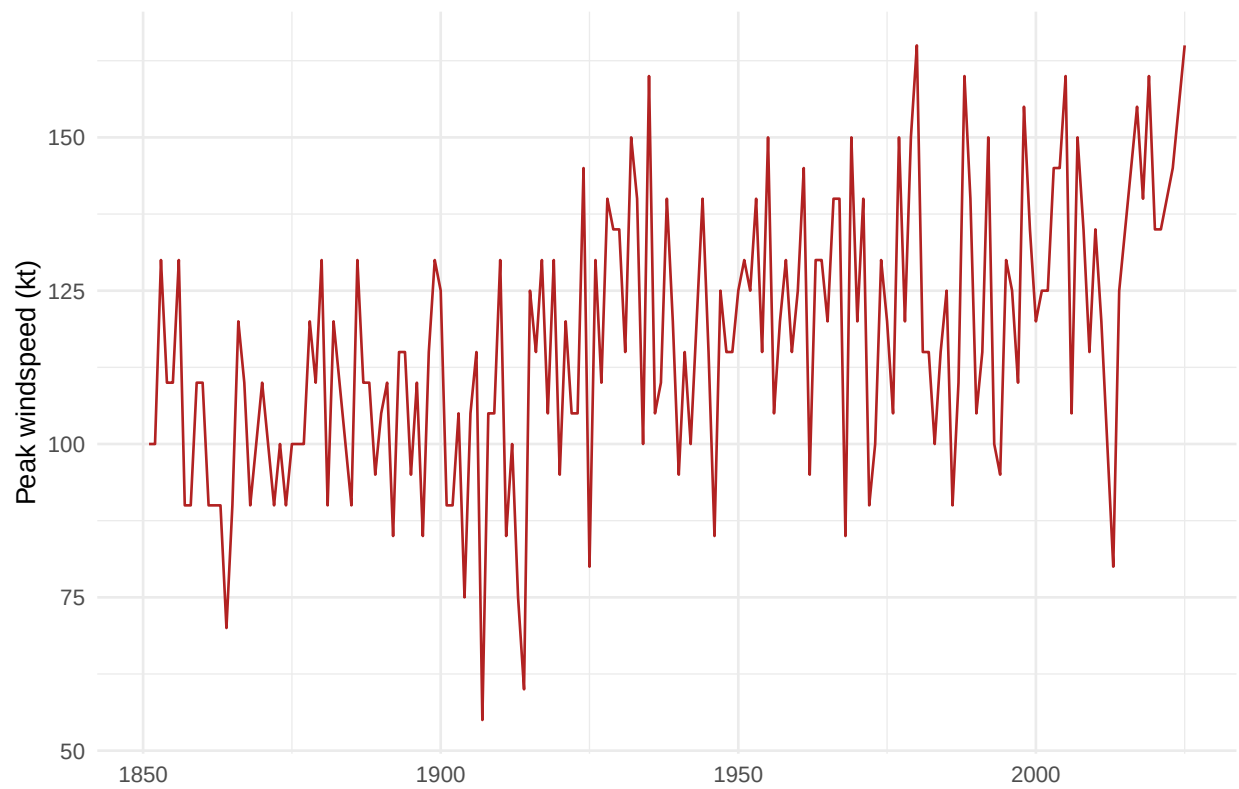


Figure 7: Annual peak windspeed. The trend is upward, but part of the pre-1950 climb reflects the gradual filling-in of missing wind reports rather than a true physical change.

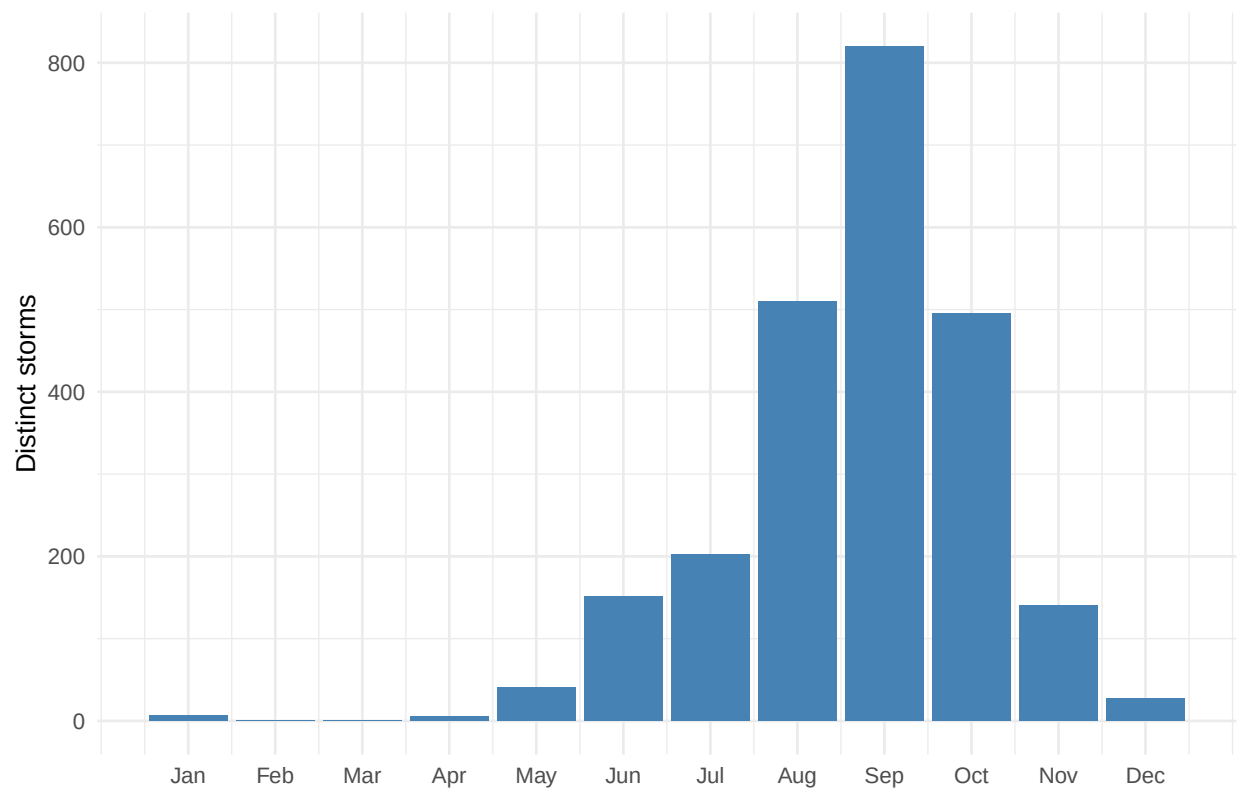


Figure 8: Distinct storms per calendar month. September is the peak of the Atlantic hurricane season.

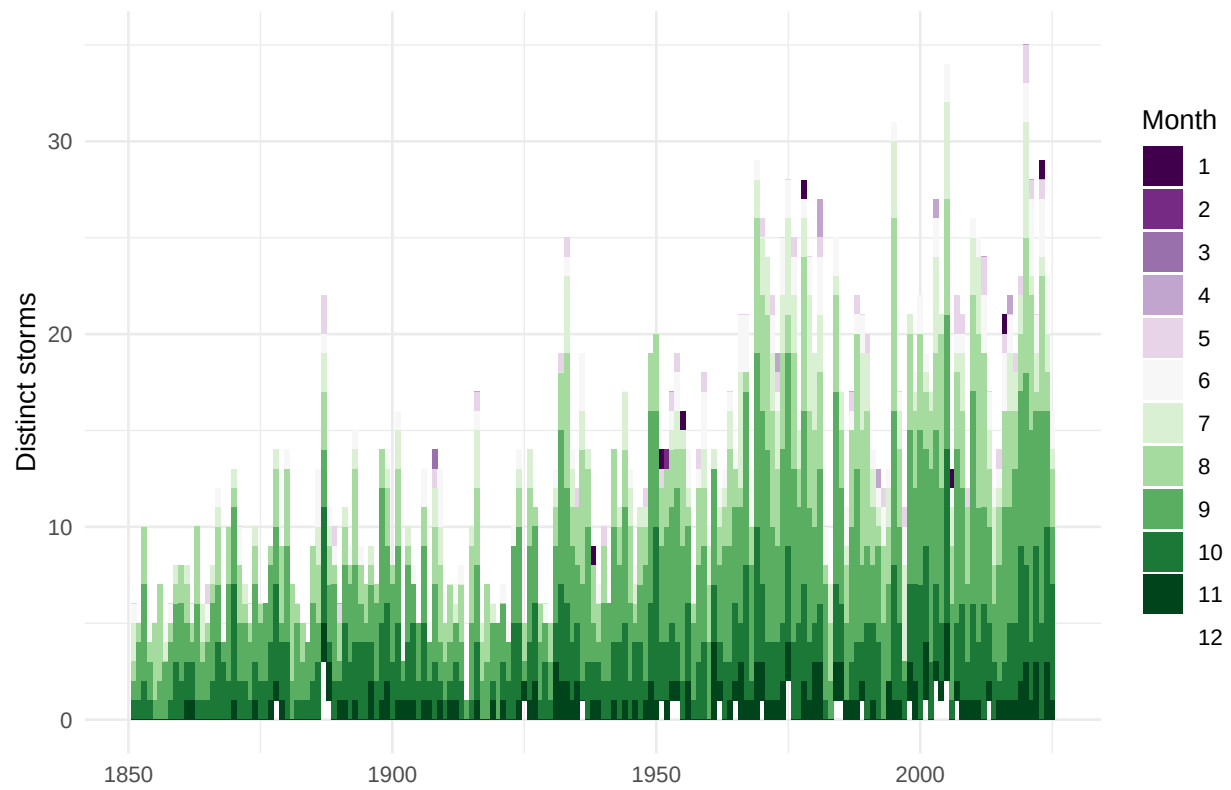


Figure 9: Storms per year, stacked by month. The coloured bands broaden into May and November-December over time.

- HURDAT2 is enormous but inhomogeneous: reporting density changes sharply across eras. Any trend in storm counts or peak windspeed has to be read against that backdrop.
- The Atlantic hurricane season is clearly concentrated in September, but the season appears to have widened over the past several decades — more storms recorded in the shoulder months of May and November-December.
- The landfall footprint has expanded poleward and westward compared with the early record. Some of that is observation; some is consistent with research on poleward migration of tropical-cyclone maximum intensity.

Reproducibility

Refresh the data with `Rscript data/fetch_data.R`. Re-render the report with `rmarkdown::render("report/hurricane-an`